STA 2002
Final Review

Outline

- **Point Estimation: MLE, MoM, Plug-in estimation**
- **Confidence Interval**
- **Testing Statistical Hypothesis**
- **Linear Regression**

Details regarding Midterm

- **Date & Time: 13:30 - 15:30 December 17, 2024 (Tuesday)**

- **Venue: TBD**
- **Allowed Items:**
  - **Two double-sided A4 cheating sheets (handwritten or typed)**
  - **A non-programmable calculator**
  - **Student ID**

Useful Results for Normal and Chi-square

- If $x_1 \sim N(\mu_1, \sigma_1^2), x_2 \sim N(\mu_2, \sigma_2^2)$ are independently normally distributed variables, then

$$y = x_1 + x_2$$

also follows the normal distribution:

$$y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

If X is $N(\mu, \sigma^2)$, then $Z = (X - \mu)/\sigma$ is $N(0,1)$.

$$X_1, X_2, \cdots, X_n \sim N(\mu, \sigma^2) \rightarrow \bar{X} \sim N(\mu, \frac{\sigma^2}{n}) \rightarrow \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0,1)$$

Sampling distribution of sample mean is normal, when samples are normal

Useful Results for Normal and Chi-square

If the random variable X is $N(\mu, \sigma^2)$, $\sigma^2 > 0$, then the random variable $V = (X - \mu)^2/\sigma^2 = Z^2$ is $\chi^2(1)$.

Facts:

- (connection to normal) $X_1, X_2, \ldots, X_r \sim N(0,1)$, and they are independent, then

$$X_1^2 + X_2^2 + \cdots + X_r^2 \sim \chi^2(r)$$

- (sum of chi-squares) $X \sim \chi^2(r_1), Y \sim \chi^2(r_2)$, and they are independent, then

$$X + Y \sim \chi^2(r_1 + r_2)$$

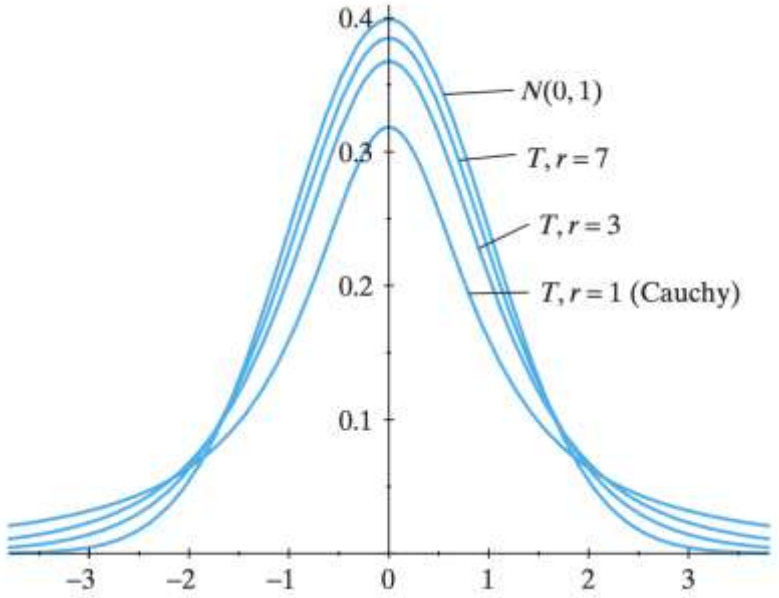
Student's t distribution

- Let $Z \sim N(0,1), U \sim \chi^2(r)$, and they are independent

$$T = \frac{Z}{\sqrt{U/r}} \sim t(r)$$: t distribution with r degrees of freedom

Facts:

The larger the degrees of freedom r, the more the t distribution looks like the normal distribution



Sample mean and variance

- Suppose we have a sequence of i.i.d. (independent and identically distributed) random variables $X_1, X_2, \ldots, X_n$. Let μ denotes their common mean, and σ^2 the common variance

- The **sample mean** is defined as

$$\overline{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

$$\mathbb{E}(\overline{X}) = \mu, \quad \mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n}$$

- The **sample variance** is defined

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X})^2$$

Sample mean and variance

If $X_1, X_2, \ldots, X_n$ are observations of a random sample of size n from the normal distribution $N(\mu, \sigma^2)$, then the distribution of the sample mean $\overline{X} = (1/n)\sum_{i=1}^{n} X_i$ is $N(\mu, \sigma^2/n)$.

Let $X_1, X_2, \ldots, X_n$ be observations of a random sample of size n from the normal distribution $N(\mu, \sigma^2)$. Then the sample mean,

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i,$$

and the sample variance,

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n} (X_i - \overline{X})^2,$$

are independent and

$$\frac{(n-1)S^2}{\sigma^2} = \frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{\sigma^2} \text{ is } \chi^2(n-1).$$

Sample mean and variance

- As a consequence, we have

$$U = \sum_{i=1}^{n} \frac{(X_i - \mu)^2}{\sigma^2} \sim \chi^2(n) \qquad W = \sum_{i=1}^{n} \frac{(X_i - \overline{X})^2}{\sigma^2} \sim \chi^2(n-1)$$

$$T = \frac{\overbrace{\frac{\overline{X} - \mu}{\sigma/\sqrt{n}}}^{N(0,1)}}{\sqrt{\frac{(n-1)S^2}{\sigma^2} \Big/ (n-1)}} = \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

$\chi^2(n-1)$

Convergence in Distribution and Probability

Definition 4.1 (Convergence in probability). *Suppose there is a sequence of random variables* $X_n, n = 1, 2, \cdots$ *and a random variable* X. *If for* $\forall \epsilon > 0$, $\exists$ *a constant* $k > 0$ *such that for any* $n \geq k$,

$$\mathbf{P}\left(|X_n - X| > \epsilon\right) \leq \epsilon, \tag{9}$$

then we say X_n *converges to* X *in probability, i.e.,* $X_n \to_p X$.

Definition 4.2 (Convergence in distribution). *Suppose we have a sequence of random variables* $X_n, n = 1, \cdots$ *and a random variable* X. *Suppose for any bounded continuous function* $h(\cdot)$

$$\mathbf{E}[h(X_n)] \to \mathbf{E}[h(X)] \quad \textit{i.e., numerically convergence} \tag{13}$$

then we say that X_n *converges to* X *in distribution, i.e.,* $X_n \to_d X$.

Law of Large Numbers (LLN)

Theorem 5.1. *Suppose random variables* $X_i, i = 1, \cdots$ *are i.i.d. (i.e., independent and identically distributed) with mean* $\mu = \mathbf{E}[X_1]$ *and* $\sigma^2 = \mathrm{Var}(X_1) < \infty$. *Define* $Z_n = \frac{1}{n}\sum_{i=1}^{n} X_i$, *then*

$$Z_n \to_p \mu. \tag{16}$$

Sample average converges to the population mean.

Central Limit Theorem (CLT)

(Central Limit Theorem) If $\overline{X}$ is the mean of a random sample $X_1, X_2, \ldots, X_n$ of size n from a distribution with a finite mean μ and a finite positive variance σ^2, then the distribution of

$$W = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sqrt{n}\,\sigma}$$

is $N(0,1)$ in the limit as $n \to \infty$.

Sampling distribution of sample mean is approximately normal, even when samples are NOT normal

Rule of thumb: when n > 30 this works pretty well.

Maximum Likelihood

Method of Moments

POINT ESTIMATOR

Estimator

Suppose X is a random variable with $f(x;\theta)$ as the pdf. If $X_1, X_2, \ldots X_n$ is a random sample of size n from X, the statistic

$$\hat{\Theta} = h(X_1, X_2, \ldots, X_n)$$

Is called a **point estimator** of θ.

After the sample has been selected, $\hat{\Theta}$ takes on a particular numerical value called the **point estimate** of θ.

Note that $\hat{\Theta}$ is a random variable because it is a statistic (function of random variables)

Unbiased Estimator

The point estimator $\hat{\Theta}$ is an **unbiased estimator** for the parameter θ if

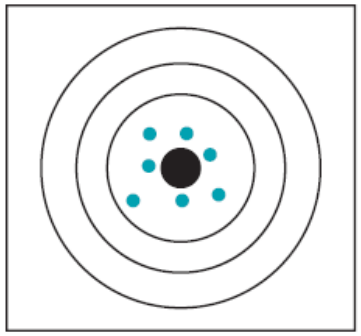
$$E(\hat{\Theta}) = \theta$$

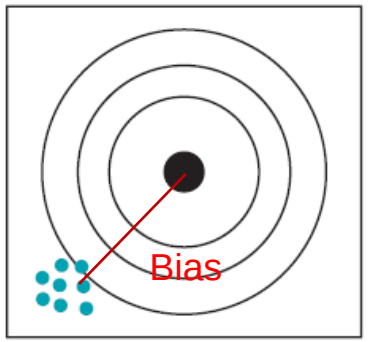
ties to sampling distribution

If the estimator is not unbiased, then the difference

$$E(\hat{\Theta}) - \theta$$

is called the **bias** of the estimator $\hat{\Theta}$.





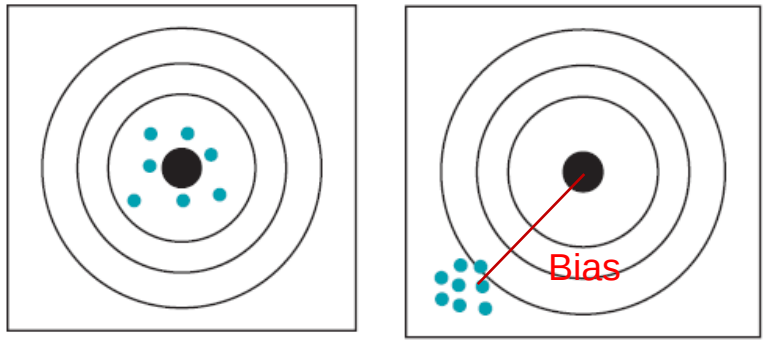
Mean Square Error (MSE)

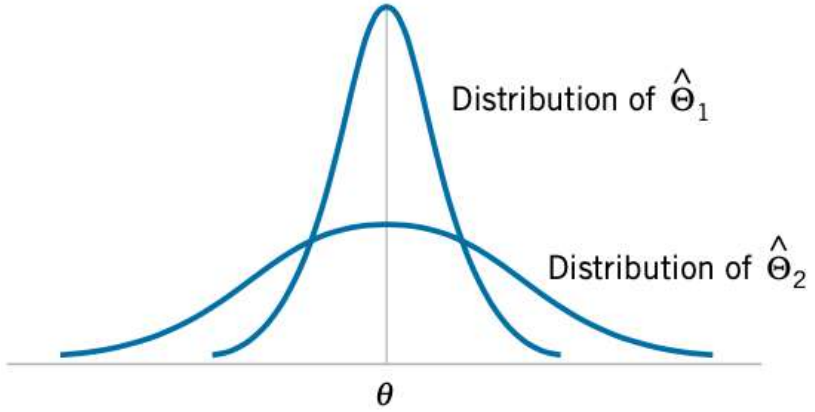
The **mean square error** of an estimator $\hat{\Theta}$ of the parameter θ is defined as

$$\text{MSE}(\hat{\Theta}) = E(\hat{\Theta} - \theta)^2$$

$$MSE(\hat{\Theta}) = E(\hat{\Theta} - \Theta)^2 = \left[E(\hat{\Theta} - \Theta)\right]^2 + \text{var}(\hat{\Theta} - \Theta)$$

$$MSE(\hat{\Theta}) = \left[\text{Bias}(\hat{\Theta})\right]^2 + \text{var}(\hat{\Theta})$$





Methods for Finding Estimators

- **Assume a distribution for the samples**
- **Estimate the parameter of the distribution**
- **Several methods**
  - **Maximum likelihood**
  - **Method of moment**

Method of Maximum Likelihood

Suppose that X is a random variable with probability distribution $f(x; \theta)$, where θ is a single unknown parameter. Let $x_1, x_2, \ldots, x_n$ be the observed values in a random sample of size n. Then the **likelihood function** of the sample is

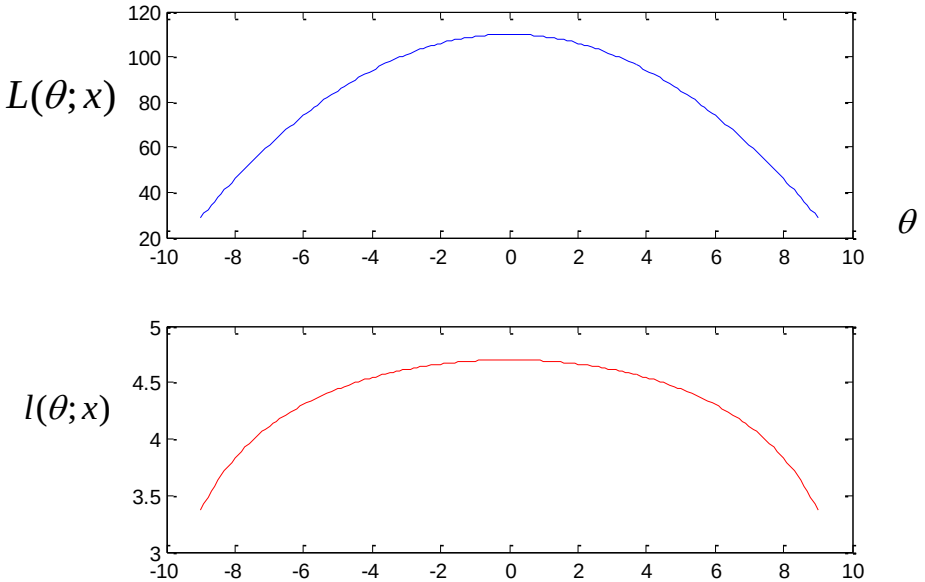
$$L(\theta) = f(x_1; \theta) \cdot f(x_2; \theta) \cdot \cdots \cdot f(x_n; \theta) \qquad (7\text{-}5)$$

Note that the likelihood function is now a function of only the unknown parameter θ. The **maximum likelihood estimator** of θ is the value of θ that maximizes the likelihood function $L(\theta)$.

$$L(\theta; x) = \prod_{i=1}^{n} f(x_i; \theta) = f(x_1; \theta) \ldots f(x_n; \theta)$$

$$l(\theta; x) = \sum_{i=1}^{n} \log\left[f(x_i; \theta) \right]$$

$$\hat{\Theta}(x) = \arg\max_{\theta} L(\theta; x) = \arg\max_{\theta} l(\theta; x)$$



Example: Normal

Let X be normally distributed with unknown μ and known variance σ^2. The likelihood function of a random sample of size n, say $X_1, X_2, \ldots, X_n$, is

$$L(\mu) = \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} e^{-(x_i-\mu)^2/(2\sigma^2)} = \frac{1}{(2\pi\sigma^2)^{n/2}} e^{-(1/2\sigma^2)\sum_{i=1}^{n}(x_i-\mu)^2}$$

Now

$$\ln L(\mu) = -(n/2)\ln(2\pi\sigma^2) - (2\sigma^2)^{-1}\sum_{i=1}^{n}(x_i-\mu)^2$$

and

$$\frac{d \ln L(\mu)}{d\mu} = (\sigma^2)^{-1}\sum_{i=1}^{n}(x_i-\mu)$$



Check second-order derivative is negative:

$\longrightarrow$ What is the MLE for μ?

$$\ell''(\hat{\mu}) = -\frac{n}{\sigma^2} < 0$$

Example (Continued, unknown variance)

$$\ln L(\mu, \sigma^2) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

$$\frac{\partial \ln L(\mu, \sigma^2)}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu) = 0$$

$$\frac{\partial \ln L(\mu, \sigma^2)}{\partial (\sigma^2)} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i - \mu)^2 = 0$$

The solutions to the above equation yield the maximum likelihood estimators

$$\hat{\mu} = \overline{X} \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

MLE: Uniform

Let X be a Uniform random variable on the interval $[0, \theta]$

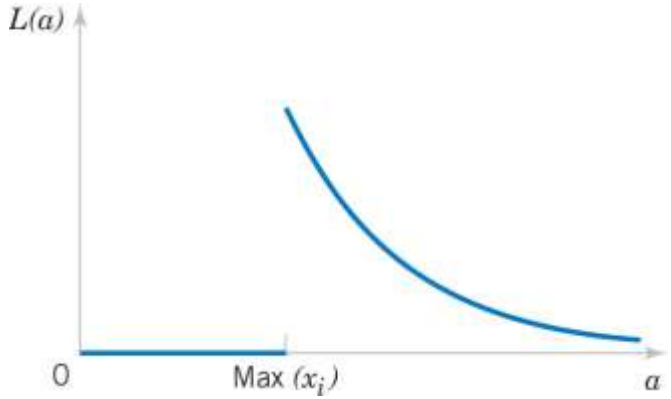
$$f(x;\theta) = \begin{cases} \frac{1}{\theta}, & \text{for } 0 \le x \le \theta, \\ 0, & \text{otherwise,} \end{cases} = \frac{1}{\theta}\mathbf{1}_{\{0 \le x \le \theta\}}$$

indicator function $\mathbf{1}_A(x)$

$$\mathbf{1}_A(x) = \begin{cases} 1, & \text{if } x \in A, \\ 0, & \text{otherwise} \end{cases}$$

The likelihood function of a random sample of size n is:

$$L(\theta) = \prod_{i=1}^{n} f(x_i;\theta) = \frac{1}{\theta^n}\prod_{i=1}^{n}\mathbf{1}_{\{0 \le x_i \le \theta\}} = \begin{cases} \frac{1}{\theta^n}, & \text{if } \theta \ge x_{(n)}, \\ 0, & \text{if } \theta < x_{(n)}. \end{cases}$$



$$\hat{\theta} = X_{(n)} = max\{X_1, X_2, \ldots, X_n\}$$

Calculus methods don't work here because $L(\theta)$ is maximized at the discontinuity. Clearly, θ cannot be smaller than max(x_i), thus the MLE is max(x_i)= $X_{(n)}$.

MLE: Uniform (continued)

Is the MLE of θ an unbiased estimator?



CDF of $X_{(n)}$: for any $x \in (0, \theta)$

$$F(x) = P\left(X_{(n)} \leq x\right) = \prod_{i=1}^{n} P(X_i \leq x) = \left(\frac{x}{\theta}\right)^n$$

PDF of $X_{(n)}$: for any $x \in (0, \theta)$

$$f(x) = F'(x) = \frac{n}{\theta}\left(\frac{x}{\theta}\right)^{n-1}$$

Mean of $X_{(n)}$:

$$E\left[X_{(n)}\right] = \int_0^\theta x f(x) dx = \int_0^\theta \frac{nx}{\theta}\left(\frac{x}{\theta}\right)^{n-1} dx = \frac{n}{\theta^n}\int_0^\theta x^n dx = \frac{n}{n+1}\theta \neq \theta$$

Variance of MLE

Fisher information

Score function $U(\theta) = \frac{\partial \ell(\theta)}{\partial \theta}$. ← log-likelihood function

Fisher $I_{1}(\theta) = \mathbf{E}\left(\frac{\partial \ell(\theta)}{\partial \theta}\right)^2 = -\mathbf{E}\left[\frac{\partial^2}{\partial \theta^2} \log(f(X_1, \ldots, X_n, \theta))\right]$

$$\sqrt{I_{1}(\theta)}(\widehat{\theta} - \theta) \to N(0, 1),$$

MLE

Equivariance of Maximum likelihood estimators

Theorem 3.1 (Equivariance of MLE). *Let $\tau = g(\theta)$ be a function of θ. Let $\hat{\theta}$ be the MLE of θ. Then, $\hat{\tau} = g(\hat{\theta})$ is the MLE of τ.*

Why MLE is optimal

Theorem 3.1 (Cramér–Rao lower bound). *Suppose θ is an unknown deterministic parameter that is to be estimated from n independent observations (measurements) of X, each from a distribution according to some probability density function $f(x;\theta)$. The variance of any unbiased estimator $\hat{\theta}$ of θ is lower bounded by the reciprocal of the Fisher information:*

$$\operatorname{Var}(\hat{\theta}) \geq \frac{1}{I_{11}(\theta)}.$$

Method of Moments

- **Equating empirical moments to theoretical moments**

Let $X_1, X_2, \ldots, X_n$ be a random sample from either a probability mass function or probability density function with m unknown parameters $\theta_1, \theta_2, \ldots, \theta_m$. The **moment estimators** $\hat{\Theta}_1, \hat{\Theta}_2, \ldots, \hat{\Theta}_m$ are found by equating the first m population moments to the first m sample moments and solving the resulting equations for the unknown parameters.

***m* equations to estimate *m* parameters**

$$\begin{cases} m_1' = \mu_1' \\ m_2' = \mu_2' \\ \vdots \\ m_m' = \mu_m' \end{cases}$$

Population moments

$$\mu_k' = \begin{cases} \int_x x^k f(x)dx & \text{continuous} \\ \sum_x x^k f(x) & \text{discrete} \end{cases}$$

Sample moments

$$m_k' = \frac{\sum_{i=1}^{n} X_i^k}{n}$$

Algebraic operations of MoM

Theorem 2.2 (CMT for arithmetic operations). *Let X_n, X, Y_n, Y be random variables, and c be some constant. Then*

- *If $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} Y$, then $X_n + Y_n \xrightarrow{p} X + Y$.*
- *If $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} Y$, then $X_n Y_n \xrightarrow{p} XY$.*
- *If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} c$, then $X_n + Y_n \xrightarrow{d} X + c$. (Slutzky's theorem)*
- *If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} c$, then $X_n Y_n \xrightarrow{d} cX$. (Slutzky's theorem)*

and $g(\cdot)$ is continuous. Then, the method of moments estimator

$$\hat{\theta} = g\left(\frac{1}{n}\sum_{i=1}^{n} X_i, \frac{1}{n}\sum_{i=1}^{n} X_i^2, ..., \frac{1}{n}\sum_{i=1}^{n} X_i^p\right) \xrightarrow{p} \theta_0.$$

Example

MoM estimator for exponential distribution?

$$X \sim Exp\left(\frac{1}{\theta}\right) \qquad f(x) = \frac{1}{\theta} e^{-x/\theta}, \qquad 0 \leq x < \infty$$

Population mean $\mu = \theta$
Sample mean = $\bar{X}$



MoM estimator: $\hat{\theta} = \bar{X}$

Unbiased

Plug-in estimation

Suppose we have a consistent estimator $\hat{\theta}$ for θ, we can *plug in* the estimator $\hat{\theta}$ to the continuous function $g(\cdot)$, and then the estimator for η,

$$\hat{\eta} := g(\hat{\theta}),$$

Two-sided vs. one-sided

CI for mean

CI for mean differences

CI for proportion

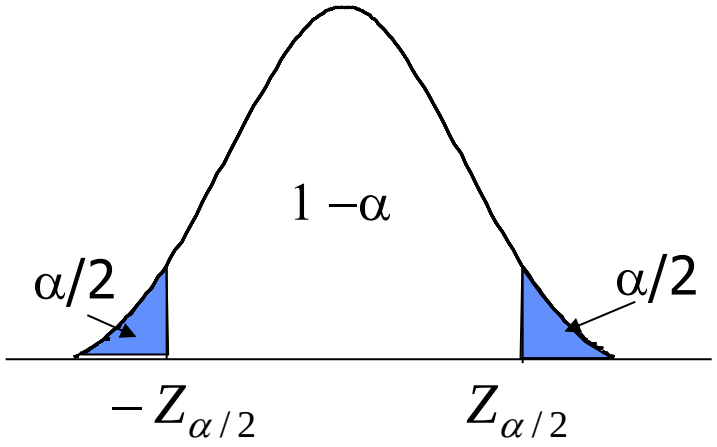
CI for MLE

CONFIDENCE INTERVAL

100(1-α)% CI for Mean of a Normal Distribution (two-sided, known variance)

- **For mean μ from a normal population with known σ,**

$$X_1, X_2, \cdots, X_n \sim N(\mu, \sigma^2) \rightarrow \bar{X} \sim N(\mu, \frac{\sigma^2}{n}) \quad \Rightarrow \quad \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$



$$P\left\{-z_{\alpha/2} \le \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \le z_{\alpha/2}\right\} = 1 - \alpha$$

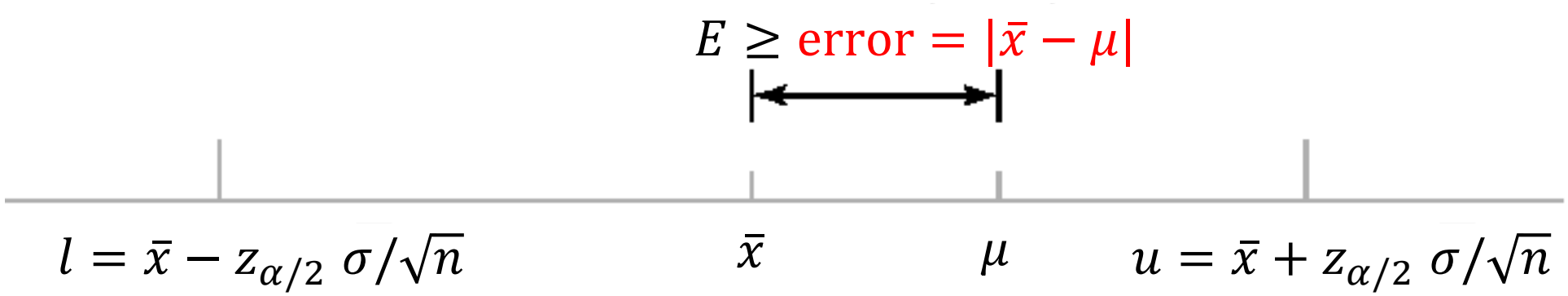
Confidence level

CI Lower CI Upper

$$\bar{x} - Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} \le \mu \le \bar{x} + Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

This CI can be used for mean of non-normal distributions (CLT)

Choice of Sample Size



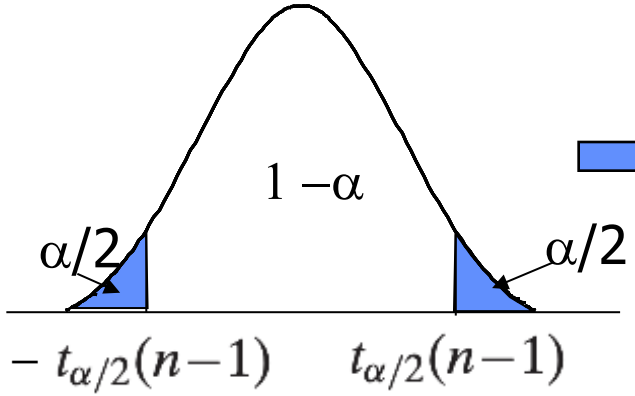
If $\bar{x}$ is used as an estimate of μ, we can be $100(1-\alpha)\%$ confident that the error $|\bar{x} - \mu|$ will not exceed a specified amount E when the sample size is

$$n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2$$

width / length of the CI = $\mathbf{2}\, z_{\alpha/2}\, \sigma/\sqrt{n}$

100(1-α)% CI for Mean of a Normal Distribution (two-sided, unknown variance)

$$X_1, X_2, \cdots, X_n \sim N(\mu, \sigma^2) \rightarrow \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$



$$P\left(-t_{\alpha/2}(n-1) \leq \frac{\overline{X} - \mu}{S/\sqrt{n}} \leq t_{\alpha/2}(n-1)\right) = 1 - \alpha$$

$$\overline{X} - t_{\alpha/2}(n-1)\left(\frac{S}{\sqrt{n}}\right) \leq \mu \leq \overline{X} + t_{\alpha/2}(n-1)\left(\frac{S}{\sqrt{n}}\right)$$

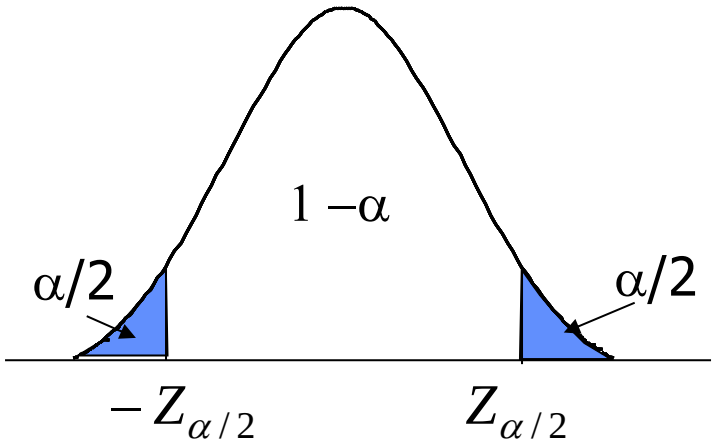
Note that $t_{\alpha/2}(n-1) \approx z_{\alpha/2}$ for *n* >= 30

This CI can be used as approximate CI for mean of non-normal distributions with unknown variance

100(1-α)% CI for a Population Proportion with large sample size

$$X_1, X_2, \cdots, X_n \sim Bernoulli(p)$$

- Using CLT, we can estimate a CI for the population proportion.



$$\hat{p} = \frac{\sum_{i=1}^{n} X_i}{n} \sim N(p, \frac{p(1-p)}{n})$$

$$\hat{p} - Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p \le \hat{p} + Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

(Approximate) Two-sided CI

Choice of Sample Size

The sample size for a specified value *E* is given by

$$n = \left(\frac{z_{\alpha/2}}{E}\right)^2 p(1-p)$$

An upper bound on *n* is given by (i.e., at least (1-α)100% confidence)

$$n = \left(\frac{z_{\alpha/2}}{E}\right)^2 0.25$$

Page 42:

If $\bar{x}$ is used as an estimate of μ, we can be $100(1-\alpha)\%$ confident that the error $|\bar{x}-\mu|$ will not exceed a specified amount E when the sample size is

$$n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2$$

Confidence Intervals for Two-Samples

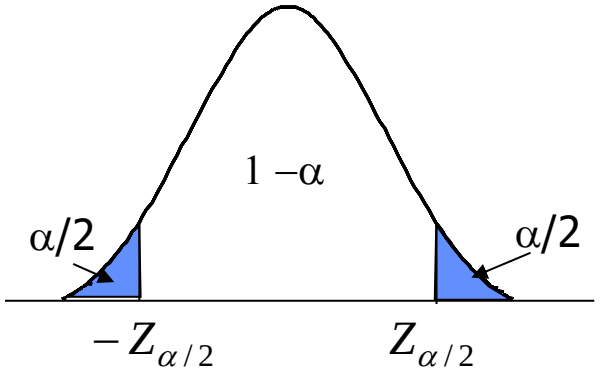
100(1-α)% CI for Differences in Means of Two Normal Distributions (two-sided, known variances)

Two independent random samples from two Normal distributions with the known variances

$$X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$$

$$Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$$

$$\Rightarrow \bar{X} - \bar{Y} \sim N\left(\mu_X - \mu_Y, \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}\right) \Rightarrow \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}} \sim N(0,1)$$



$$P\left\{-z_{\alpha/2} \leq \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}} \leq z_{\alpha/2}\right\} = 1 - \alpha$$

← **Confidence level**

CI Lower **CI Upper**

$$\bar{X} - \bar{Y} - z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}} \leq \mu_X - \mu_Y \leq \bar{X} - \bar{Y} + z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$$

This CI can be used for mean of non-normal distributions when *n*,*m*>30

100(1-α)% CI for Difference in Means of Two Normal Distributions (two-sided, unknown but equal variances)

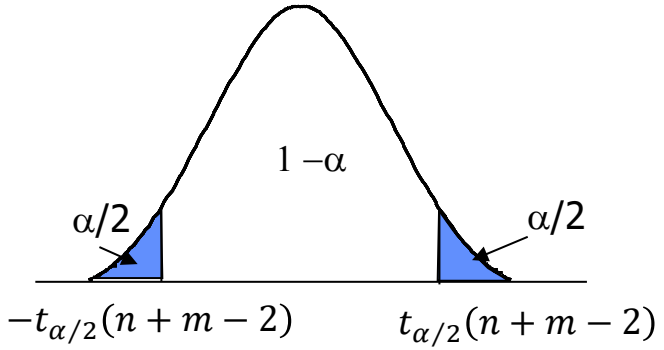
Two independent random samples from two Normal distributions with unknown (but equal) variances

$$X_1, \ldots, X_n \sim N(\mu_X, \sigma^2)$$

$$Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma^2)$$

$$\Rightarrow \quad \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \sim t(n + m - 2)$$

$$S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n + m - 2}$$



$$\Rightarrow \quad P\left\{-t_{\alpha/2}(n+m-2) \le \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \le t_{\alpha/2}(n+m-2)\right\} = 1 - \alpha$$

$$\bar{X} - \bar{Y} - t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}} \le \mu_X - \mu_Y \le \bar{X} - \bar{Y} + t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$$

Two-sided

100(1-α)% CI for Difference in Population Proportions with large sample size

$$\left.\begin{array}{l} X_1, X_2, \cdots, X_{n_1} \sim Bernoulli(p_1) \\ Y_1, Y_2, \cdots, Y_{n_2} \sim Bernoulli(p_2) \end{array}\right\} \rightarrow \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}} \sim N(0,1)$$

For large sample sizes

Two-sided CI

$$(\hat{p}_1 - \hat{p}_2) - Z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \leq (p_1 - p_2) \leq (\hat{p}_1 - \hat{p}_2) + Z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

CI for MLE

In order to construct the (two-sided) confidence interval, we first need a consistent estimator for the Fisher information, i.e., find an estimator $\widehat{I}_1(\theta)$ such that

$$\widehat{I}_1(\theta) \to_p I_1(\theta)$$

$$\text{then } \sqrt{n \times \widehat{I}_1(\theta)} \times (\widehat{\theta} - \theta) \to_d N(0,1)$$

Thus $\sqrt{n \times \widehat{I}_1(\theta)} \times (\widehat{\theta} - \theta)$ is a pivot.

To establish the confidence interval, suppose $z_{\alpha/2}$ still being the upper $\alpha/2$ quantile of a standard normal random variable, define the confidence interval

$$\left[\widehat{\theta} - \frac{z_{\frac{\alpha}{2}}}{\sqrt{n\widehat{I}_1(\theta)}}, \quad \widehat{\theta} + \frac{z_{\frac{\alpha}{2}}}{\sqrt{n\widehat{I}_1(\theta)}}\right]$$

Recap:Summary of Sampling Distributions

		One population	Two populations
Sample Mean	Known Variance	$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$	$\frac{\bar{X}-\bar{Y}-(\mu_X-\mu_Y)}{\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}} \sim N(0,1)$
	Unknown Variance	$\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t(n-1)$	$\frac{\bar{X}-\bar{Y}-(\mu_X-\mu_Y)}{S_p\sqrt{\frac{1}{n}+\frac{1}{m}}} \sim t(n+m-2)$
Sample proportion		$\hat{p} \sim N(p, p(1-p)/n)$	$\frac{(\hat{p}_1-\hat{p}_2)-(p_1-p_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}+\frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}} \sim N(0,1)$
Sample Variance		$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$	$\frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \sim F(n-1, m-1)$

Summary

		One population	Two populations
Sample Mean	Known Variance	$\bar{x} - z_{\alpha/2}\sigma/\sqrt{n} \le \mu \le \bar{x} + z_{\alpha/2}\sigma/\sqrt{n}$ $\mu \ge \bar{X} - Z_\alpha \times \frac{\sigma}{\sqrt{n}}$ $\quad$ $\mu \le \bar{X} + Z_\alpha \times \frac{\sigma}{\sqrt{n}}$	$\bar{X} - \bar{Y} - z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}} \le \mu_X - \mu_Y \le \bar{X} - \bar{Y} + z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$ $\bar{X} - \bar{Y} - z_\alpha\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}} \le \mu_X - \mu_Y$ $\quad$ $\mu_X - \mu_Y \le \bar{X} - \bar{Y} + z_\alpha\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$
	Unknown Variance	$\bar{X} - t_{\alpha/2}(n-1)\left(\frac{S}{\sqrt{n}}\right) \le \mu \le \bar{X} + t_{\alpha/2}(n-1)\left(\frac{S}{\sqrt{n}}\right)$ $\mu \le \bar{x} + t_\alpha(n-1)(s/\sqrt{n})$ $\mu \ge \bar{x} - t_\alpha(n-1)(s/\sqrt{n})$	$\bar{X} - \bar{Y} - t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}} \le \mu_X - \mu_Y \le \bar{X} - \bar{Y} + t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$ $\mu_X - \mu_Y \ge \bar{X} - \bar{Y} - t_\alpha(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$ $\mu_X - \mu_Y \le \bar{X} - \bar{Y} + t_\alpha(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$
Sample proportion		$\hat{p} - Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p \le \hat{p} + Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ $\hat{p} - Z_\alpha\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p$ $\quad$ $p \le \hat{p} + Z_\alpha\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$(\hat{p}_1 - \hat{p}_2) - Z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \le (p_1 - p_2) \le (\hat{p}_1 - \hat{p}_2) + Z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$ $(\hat{p}_1 - \hat{p}_2) - Z_\alpha\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \le (p_1 - p_2)$ $\quad$ $(p_1 - p_2) \le (\hat{p}_1 - \hat{p}_2) + Z_\alpha\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$
Sample Variance		$\frac{(n-1)s^2}{\chi^2_{\alpha/2}(n-1)} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}(n-1)}$ $\sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha}(n-1)}$ $\quad$ $\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_\alpha(n-1)}$	$\frac{S_X^2}{S_Y^2}F_{1-\alpha/2}(m-1,n-1) \le \frac{\sigma_X^2}{\sigma_Y^2} \le \frac{S_X^2}{S_Y^2}F_{\alpha/2}(m-1,n-1)$ $\frac{\sigma_X^2}{\sigma_Y^2} \ge \frac{S_X^2}{S_Y^2}F_{1-\alpha}(m-1,n-1)$ $\quad$ $\frac{\sigma_X^2}{\sigma_Y^2} \le \frac{S_X^2}{S_Y^2}F_\alpha(m-1,n-1)$

Concept of Hypothesis Testing, Type-I / Type-II Error

P-value and Power

Two-sample test

Likelihood ratio test

TESTING STATISTICAL HYPOTHESIS

Hypothesis Testing Procedures

1. **Parameter of interest:** From the problem context, identify the parameter of interest
2. **Null hypothesis H_0:** State the null hypothesis – initial belief/claim/assumption
3. **Alternative hypothesis H_1:** Specify an appropriate alternative hypothesis
4. **Test statistic:** Determine an appropriate test statistic
5. **Reject H_0 if :** State the rejection criteria for the null hypothesis
6. **Computation:** Compute any necessary sample quantities, substitute these into the equation for the test statistic, and compute that value
7. **Draw conclusion:** Decide whether or not H_0 should be rejected and report that in the problem context
8. **Report p-values**

- **Two ways to perform hypothesis test**
  - **— Rejection region**
  - **— Compare p-value with α**

Errors in Hypothesis Test

Type I Error

Rejecting the null hypothesis H_0 when it is true is defined as a **type I error.**

Probability of Type I Error

$\alpha = P(\text{type I error}) = P(\text{reject } H_0 \text{ when } H_0 \text{ is true})$

Type II Error

Failing to reject the null hypothesis when it is false is defined as a **type II error.**

Probability of Type II Error

$\beta = P(\text{type II error}) = P(\text{fail to reject } H_0 \text{ when } H_0 \text{ is false})$

Decision	H_0 Is True	H_0 Is False
Fail to reject H_0	no error	type II error
Reject H_0	type I error	no error

Significance Level

- Generally, the statistician controls type-I error probability α when she selects the critical values
- Set type-I error probability at a desired level called significance level
- Deal with the corresponding type-II error β resulted from this

Typical value for significance level: 0.05, 0.1, etc.

P-value and its distribution

Definition 3.1. *Suppose we have hypothesis H_0 versus H_1 based on the test statistics T and the rejection region $R = R(\alpha)$, i.e., the rejection region depends on the significance level α. For an observed test statistics value T, the $P-$value is defined by*

$$\widehat{p} = \inf\{\alpha \in [0,1] : T \in R(\alpha)\} \tag{2}$$

P - value should have uniform distribution under H0, and should converge in probability to 0 under H1

Power of a test

Definition 4.1. *The* **Power function** $K(\theta)$ *of a hypothesis test with rejection region* R *is* $\mathbf{P}_\theta(X \in R)$.

Suppose $\beta(\theta)$ is the Type-II error, then we have $K(\theta) = 1 - \beta(\theta)$.

As an example, suppose data $X_1, X_2, \cdots X_n \sim i.i.d.\ N(\mu, \sigma)$ **and we want to test**

$$H_0 : \mu = \mu_0, versus\ H_1 : \mu > \mu_0$$

Then the power is given by

$$1 - \Phi(z_\alpha + \frac{\sqrt{n}(\mu_0 - \mu)}{\sigma})$$

Testing normal mean
(Known Variance)

Data samples

$$X_1, \ldots, X_n \sim N(\mu, \sigma^2)$$

Null Hypothesis

$$H_0 : \mu = \mu_0$$

Alternative Hypothesis

$$H_1 : \mu \neq \mu_0 \quad \text{two-sided}$$

$$H_1 : \mu > \mu_0$$

$$H_1 : \mu < \mu_0 \quad \text{one-sided}$$

Test Statistic

$$Z_0 = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

Rejection region

R:

$$|Z_0| > Z_{\alpha/2}$$

$$Z_0 > Z_\alpha$$

$$Z_0 < -Z_\alpha$$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

two-sided **Power function** $K(\mu) = 1 - \beta = P_{\mu \neq \mu_0}(Z_0 \in R) = P_{\mu \neq \mu_0}(|Z_0| > Z_{\alpha/2})$

Under $\mu > \mu_0$ **Power function** $K(\mu) = 1 - \beta = P_{\mu > \mu_0}(Z_0 \in R) = P_{\mu > \mu_0}(Z_0 > Z_\alpha)$

Under $\mu < \mu_0$ **Power function** $K(\mu) = 1 - \beta = P_{\mu < \mu_0}(Z_0 \in R) = P_{\mu < \mu_0}(Z_0 < -Z_\alpha)$

Power function, denoted by $K(\mu)$, is a function of the parameter μ.

Two sample test for normal mean

Recall the CI lecture: $X_{11}, \ldots, X_{1n_1}$ be iid samples from $N(\mu_1, \sigma_1^2)$, and $X_{21}, \ldots, X_{2n_2}$ be iid samples from $N(\mu_2, \sigma_2^2)$. First assume σ_1^2, σ_2^2 are known. Denote the sample mean

$$\bar{X}_1 = \frac{1}{n_1}\sum_{i=1}^{n_1} X_{1i}, \quad \bar{X}_2 = \frac{1}{n_2}\sum_{i=1}^{n_2} X_{2i}.$$

Then we have

$$\frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}} \sim N(0,1)$$

First consider the two-side test

$$H_0 : \mu_1 = \mu_2 \text{ and } \; H_1 : \mu_1 \neq \mu_2$$

Define the test statistic and the rejection region

$$\widehat{T} = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}} \text{ and } R_\alpha = (-\infty, -z_{\alpha/2}) \cup (z_{\alpha/2}, \infty)$$

Second, if we assume unknown but equal variance $\sigma_1^2 = \sigma_2^2 = \sigma^2$, from Lecture 3 we have the pooled estimator of σ^2:

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}.$$

and we have the following test statistics

$$\widehat{T} = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t(n_1 + n_2 - 2).$$

1. For two-sided test, the rejection region is $R_\alpha = (-\infty, -t_{\alpha/2, n_1+n_2-2}) \cup (t_{\alpha/2, n_1+n_2-2}, \infty)$, and the $P-$value equals to $\mathbf{P}(|T| \geq |\widehat{T}|)$ where $T \sim t_{n_1+n_2-2}$ and $\widehat{T}$ is the observed value of test statistics.

2. For one-sided test $(\mu_1 > \mu_2)$, the rejection region is $R_\alpha = (t_{\alpha, n_1+n_2-2}, \infty)$, and the $P-$value equals to $\mathbf{P}(T \geq \widehat{T})$.

Recall the setting from lecture 3 (Confidence Interval): $X_{11}, \ldots, X_{1n_1}$ be iid samples from $Bernoulli(n_1, p_1)$, and $X_{21}, \ldots, X_{2n_2}$ be iid samples from $Bernoulli(n_2, p_2)$. Denote

$$\hat{p}_1 = \frac{1}{n_1}\sum_{i=1}^{n_1} X_{1i}, \quad \hat{p}_2 = \frac{1}{n_2}\sum_{i=1}^{n_2} X_{2i}.$$

Then we have for moderately large sample size n,

$$\frac{\hat{p}_1 - \hat{p}_2 - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}2)}{n_2}}} \sim_{Approx} N(0,1)$$

- Consider the two-side test

$$H_0 : p_1 = p_2 \text{ and } \quad H_1 : p_1 \neq p_2$$

Define the test statistic

$$\widehat{T} = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}2)}{n_2}}}$$

Then the rejection region is

$$R_\alpha = (-\infty, -z_{\alpha/2}] \cup [z_{\alpha/2}, \infty)$$

Likelihood Ratio Test

LRT for simple null

$$H_0 : \ \theta = \theta_0, \ \text{here } \theta_0 \text{ is a known number}$$

Theorem 1.1. *Suppose data $X_1, X_2, \cdots, X_n$ are independent and generated from pmf/density $f(x;\theta)$, i.e., the density and pmf is controlled by the parameter $\theta \in \mathbb{R}$ (scalar parameter). For testing*

$$H_0 : \ \theta = \theta_0, \ \textit{here } \theta_0 \textit{ is a known number}$$

Define the test statistics

$$\widehat{T} = 2(\ell(\widehat{\theta}) - \ell(\theta_0)) \quad \textbf{(LRT)} \tag{1}$$

then under H_0,

$$\widehat{T} \to_d \chi^2_1$$

Example

Suppose data $X_1, \cdots, X_n$ are i.i.d. $N(\mu, \sigma^2)$ distributed random variables, then the log–likelihood function is

$$\ell(\mu) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2 - \frac{n}{2}\log(2\pi\sigma^2),\ \widehat{\mu} = \frac{1}{n}\sum_{i=1}^{n}X_i.$$

Therefore, the likelihood ratio test statistics for $H_0 : \mu = \mu_0$ is

$$\widehat{T} = 2(\frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu_0)^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \widehat{\mu})^2) = \frac{(\widehat{\mu} - \mu_0)^2}{\sigma^2},$$

and the p–value is $\widehat{p} = 1 - \Psi(\widehat{T})$.

Wald Test and Score Test

$$H_0 : \ \theta = \theta_0, \ \text{here } \theta_0 \text{ is a known number}$$

Wald test. $$\widehat{W} = I(\widehat{\theta})(\widehat{\theta} - \theta_0)^2 \to_d \chi_1^2$$

Notably, here I(.) refers to the population (not sample) Fisher information

score test. $$U(\theta) = \frac{\partial}{\partial \theta} \log(f(X_1, \cdots, X_n, \theta))$$

$$\widehat{T} = \frac{U^2(\theta_0)}{I(\theta_0)} \to_d \chi_1^2$$

General Likelihood Ratio Test

We generalize the above likelihood ratio type tests to general settings, where we may have **composite null hypotheses** and composite alternative hypotheses.

A likelihood ratio test (LRT) statistic for testing $H_0 : \theta \in \Theta_0$ vs. $H_1 : \theta \in \Theta_1$ is

$$\lambda(x) = \frac{\sup_{\theta \in \Theta_0} L(\theta|x)}{\sup_{\theta \in \Theta} L(\theta|x)} = \frac{\sup_{\theta \in \Theta_0} L(\theta|x)}{\max\{\sup_{\theta \in \Theta_0} L(\theta|x), \sup_{\theta \in \Theta_1} L(\theta|x)\}}$$

where $\Theta = \Theta_0 \cup \Theta_1$. A LRT accepts H_0 if $\lambda(x)$ is large enough, and rejects H_0 is $\lambda(x)$ is too small, i.e., the rejection region of LRT is $\{x : \lambda(x) \leq c\}$ for some $c \in [0, 1]$.

Theorem 2.1 (Wilks). *Suppose the dimension of Θ_0 is q and the dimension of Θ is r, define*

$$\lambda_n = 2\log\left(\frac{\sup_{\theta\in\Theta}\mathcal{L}_n(\theta)}{\sup_{\theta\in\Theta_0}\mathcal{L}_n(\theta)}\right) = 2\log\left(\frac{\mathcal{L}_n(\hat{\theta})}{\mathcal{L}_n(\hat{\theta}_0)}\right),$$

where $\hat{\theta}$ is the MLE and $\hat{\theta}_0$ is the MLE when the parameter space is constrained to Θ_0.

Then under the null hypothesis,

$$\lambda_n \xrightarrow{d} \chi^2_{r-q},$$

where χ^2_{r-q} denotes the χ^2 distribution with the degree of freedom being $r-q$.

Moreover, the p-value for the test is $\mathbf{P}(\chi^2_{r-q} \geq \lambda_n)$.

Example 4: composite null Again testing normal mean, $X_1, \ldots, X_n$ from $N(\mu, \sigma^2)$ with σ^2 known. Consider testing

$$H_0 : \mu \leq \mu_0 \text{ versus } H_1 : \mu > \mu_0$$

In such cases, we have $\Theta_0 = (-\infty, \mu_0]$ and $\Theta_1 = [\mu_0, +\infty)$, $\Theta = \mathbb{R}$. Therefore we have $\widehat{\theta}_0 = \min\{\bar{X}, \mu_0\}$, $\widehat{\theta} = \bar{X}$, and thus the likelihood ratio statistics is

$$\lambda(x) = \frac{L(\widehat{\theta}_0)}{L(\widehat{\theta})} = \begin{cases} 1 & \text{if } \bar{X} \leq \mu_0 \\ \exp\left(-n\frac{1}{2\sigma^2}(\bar{X} - \mu_0)^2\right) & \text{if } \bar{X} > \mu_0 \end{cases}$$

Therefore, the LRT will reject H_0 when $\lambda(x) \leq c$, which happens iff $\bar{X} > \mu_0$ and $\bar{X} - \mu_0 \geq k$, thus the rejection region is

$$R(\alpha) = \{\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \geq z_\alpha\}.$$

Furthermore, we note that under the null hypothesis, if the true mean value is $\mu < \mu_0$, the corresponding type-I error

$$\beta(\mu) = \mathbf{P}\{\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \geq z_\alpha\} = \mathbf{P}\{\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} + \frac{\mu - \mu_0}{\sigma/\sqrt{n}} \geq z_\alpha\} \leq \alpha$$

Which means that for composite null hypothesis, the type-I error is always controlled by α.

Linear Regression: least-square (LS) method

Definition 3.1. *We suppose a good linear model f should minimize the 'empirical square error'*

$$L(f;\{x_{i1},x_{i2},\cdots,x_{ip},y_i\}_{i=1}^n)=\frac{1}{n}\sum_{i=1}^n(y_i-f(x_{i1},x_{i2},\cdots,x_{ip}))^2,\ \ f\in\mathcal{F}.$$

The model f that minimizes the square error is called the **least-square** *fit.*

Since we assume $\mathcal{F}$ the group of linear functions, so

$$\min_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n(y_i-f(x_{i1},x_{i2},\cdots,x_{ip}))^2=\min_{\beta_1,\cdots,\beta_p\in\mathbf{R}}\frac{1}{n}\sum_{i=1}^n(y_i-\sum_{j=1}^p x_{ij}\beta_j)^2:=\min_{\beta_1,\cdots,\beta_p\in\mathbf{R}}L(\beta_1,\cdots,\beta_p)$$

By taking derivative, we have

$$\frac{\partial L}{\partial\beta_k}=0\Rightarrow\frac{1}{n}\sum_{i=1}^n x_{ik}y_i=\sum_{j=1}^p\left(\frac{1}{n}\sum_{i=1}^n x_{ij}x_{ik}\right)\times\beta_j\ \text{ for }\ k=1,2,\cdots,p.$$

Use the matrix representation in eq.(1), we have

$$\frac{1}{n}\mathbf{X}^T\mathbf{y}=(\frac{1}{n}\mathbf{X}^T\mathbf{X})\boldsymbol{\beta},\ \text{ here }\ \boldsymbol{\beta}=(\beta_1,\beta_2,\cdots,\beta_p)^T\in\mathbf{R}^{p\times1}.$$

So, if this linear equality has solution, i.e., if the matrix $\mathbf{X}^T\mathbf{X}$ has rank p (invertible), then we have

$$\widehat{\boldsymbol{\beta}}=\underset{\beta}{\operatorname{argmin}}\,L(\boldsymbol{\beta})=(\frac{1}{n}\mathbf{X}^T\mathbf{X})^{-1}\left(\frac{1}{n}\mathbf{X}^T\mathbf{y}\right)=(\mathbf{X}^T\mathbf{X})^{-1}\left(\mathbf{X}^T\mathbf{y}\right).$$

Variance Estimation

Theorem 7.1. *Suppose observations* $(x_{i1}, x_{i2}, \cdots, x_{ip}, y_i), i = 1, \cdots, n$ *satisfy*

$$y_i = \sum_{j=1}^{p} x_{ij}\beta_j + \epsilon_i, \ \ here\ \epsilon_i \sim N(0, \sigma^2)$$

then the estimator

$$\widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} \widehat{\epsilon}_i^2, \quad here\ \widehat{\epsilon}_i = y_i - \sum_{j=1}^{p} x_{ij}\widehat{\beta}_j \tag{2}$$

satisfies

$$\widehat{\sigma}^2 \to_p \sigma^2 \ \ as\ n \to \infty.$$

Maximum likelihood estimation

where $\epsilon_i \sim N(0, \sigma^2)$, $i = 1, 2, \ldots, n$. In other words, y_i satisfies normal distribution $N(\sum_{j=1}^{p} x_{ij}\beta_j, \sigma^2)$.

We can also apply the maximum likelihood method to fit $\boldsymbol{\beta}$. To do that, note that

$$y_i = \sum_{j=1}^{p} x_{ij}\beta_j + \epsilon_i \sim N(\sum_{j=1}^{p} x_{ij}\beta_j, \sigma^2)$$

$$\Rightarrow \text{Likelihood } f(y_1, \cdots y_n) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \sum_{j=1}^{p} x_{ij}\beta_j)^2\right)$$

$$\Rightarrow \text{ Log-likelihood } \ell(\beta_1, \cdots, \beta_p, \sigma^2) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \sum_{j=1}^{p} x_{ij}\beta_j)^2 - \frac{n}{2}\log(2\pi\sigma^2)$$

Taking the gradient, we have

$$\frac{\partial L}{\partial \beta_k} = 0 \Rightarrow \sum_{i=1}^{n}(y_i - \sum_{j=1}^{p} x_{ij}\beta_j) \times (-x_{ik}) = 0$$

$$\frac{\partial L}{\partial (\sigma^2)} = 0 \Rightarrow \sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \sum_{j=1}^{n} x_{ij}\beta_j)^2$$

Therefore, the maximum-likelihood estimator of β, σ^2 is

$$\widehat{\beta} = (\mathbf{X^T X})^{-1}\mathbf{X^T y}, \text{ and } \widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \sum_{j=1}^{p} x_{ij}\widehat{\beta}_j)^2$$

Fitting a linear model with 1 regressor

The alternative matrix form: In this situation, we have from Section 3 above:

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \Rightarrow \mathbf{X}^T\mathbf{X} = \begin{bmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{bmatrix} \text{ and } \mathbf{X}^T\mathbf{y} = \begin{bmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n x_i y_i \end{bmatrix}$$

Then

$$(\mathbf{X}^T\mathbf{X})^{-1} = \begin{bmatrix} n & n\bar{x} \\ n\bar{x} & T_{xx} \end{bmatrix}^{-1} = \frac{1}{nT_{xx} - n^2\bar{x}^2} \begin{bmatrix} T_{xx} & -n\bar{x} \\ -n\bar{x} & n \end{bmatrix} = \frac{1}{nS_{xx}} \begin{bmatrix} T_{xx} & -n\bar{x} \\ -n\bar{x} & n \end{bmatrix}$$

and thus

$$\begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \end{bmatrix} = \frac{1}{S_{xx}} \begin{bmatrix} T_{xx} & -n\bar{x} \\ -n\bar{x} & n \end{bmatrix} \begin{bmatrix} \bar{y} \\ \frac{1}{n}T_{xy} \end{bmatrix} = \frac{1}{S_{xx}} \begin{bmatrix} T_{xx}\bar{y} - T_{xy}\bar{x} \\ T_{xy} - n\bar{x}\bar{y} \end{bmatrix}$$

Which means that the

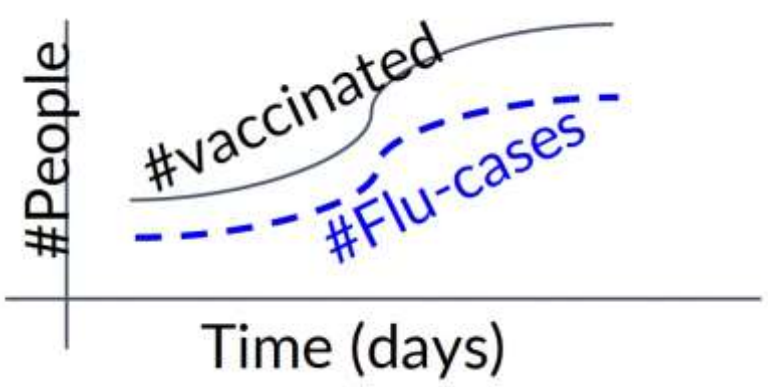
$$\widehat{\beta}_1 = \frac{T_{xy} - n\bar{x}\bar{y}}{S_{xx}} = \frac{S_{xy}}{S_{xx}}, \quad \widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1\bar{x}.$$

However, as you can see, it is very complex for $p > 1$ without matrix representation, so we recommend using matrix representation when you try to analyze the linear model.

Possible course of interests for future study

STA4041 Causal Inference

- **Correlation ≠ Causation, but how to discover the causal**





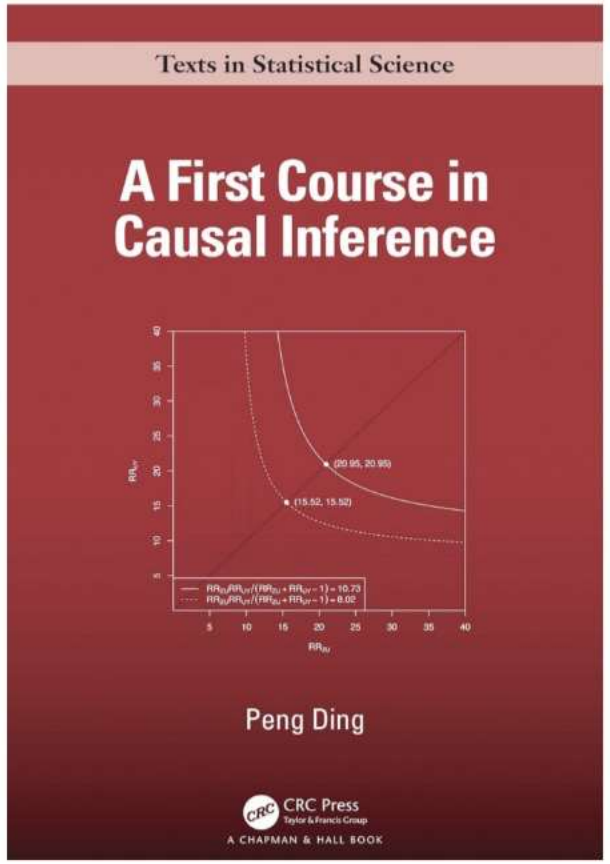
- **Causal Analysis is Important, for example:**



Clinical Trial (COVID vaccine!)



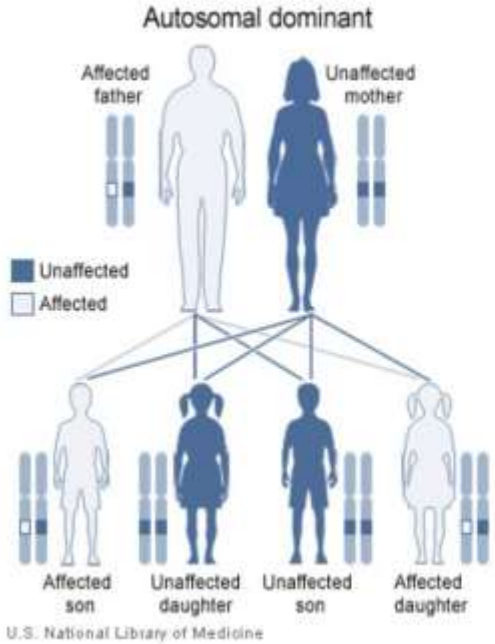
Does quitting smoking reduce insurance premium?



We will systemically explore how to perform causal analysis in this course.

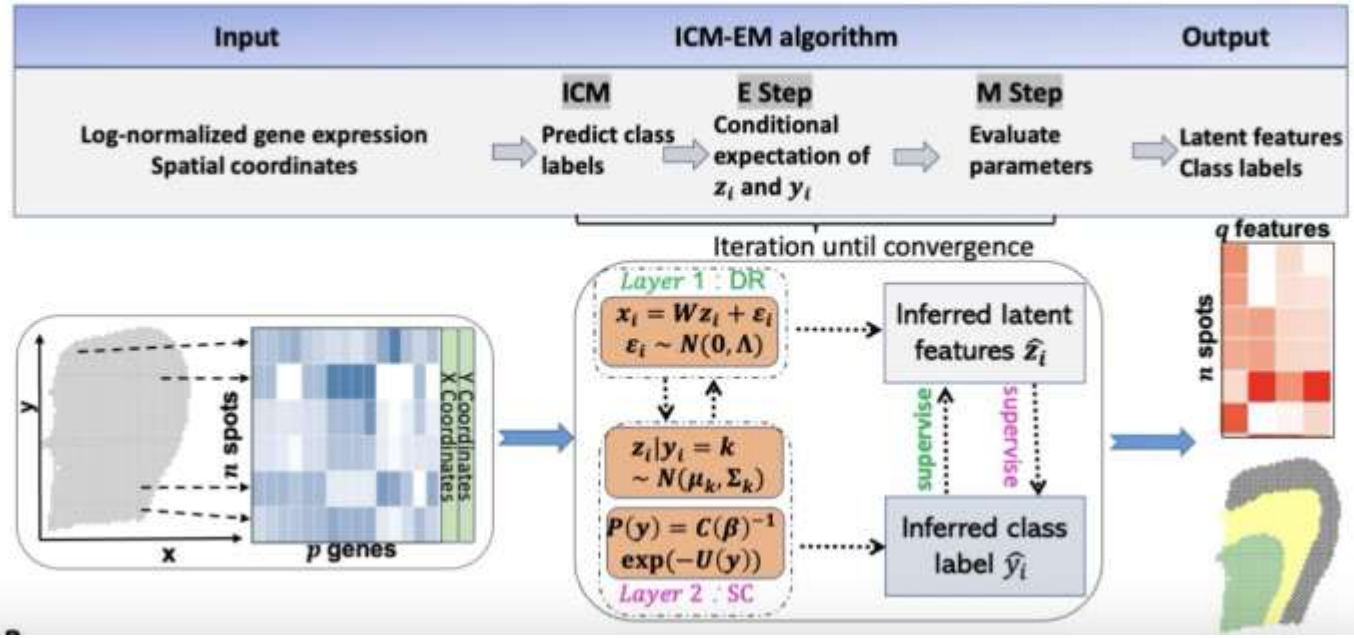
DDA4010: Bayesian Statstics

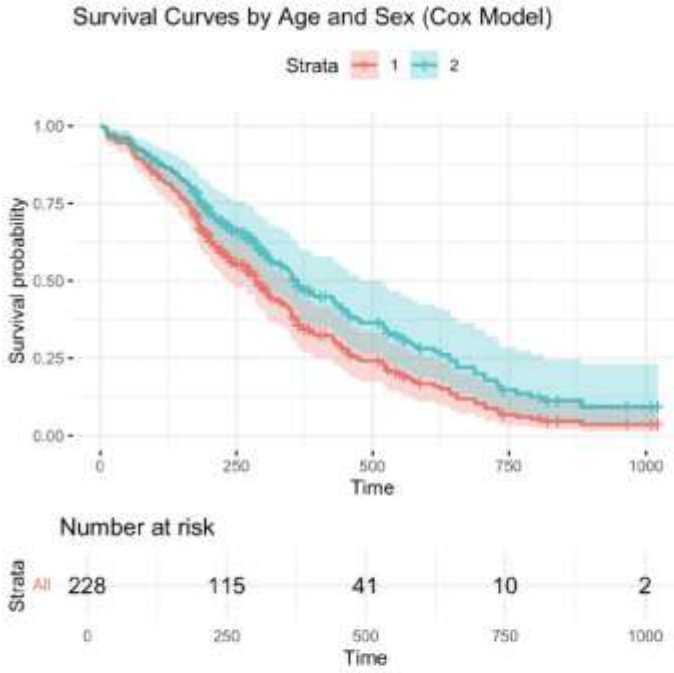
- Most statistics courses you have learned introduce **frequentist** approches (parameters are fixed value, while only estimators are random)
- **Bayesian** approaches directly quantify the **uncertainty** of parameters by prior (before observing data) and posterior (after observing data) distributions
- Advantages in solving
  - Small sample problems
  - Hierachical modeling: flexible models fitting real-world data well
- Widely used in various fields:
  - Personalized medicine in healthcare
  - Risk assessment and portfolio optimization in finance
  - Natural language processing and computer vision in deep learning
  - Climate prediction in environmental science
- This course will emphasize both **conceptual foundations and computational strategies**, including conjugate priors, Markov Chain Monte Carlo, Bayes factors, missing data imputation, Bayesian regression, hierarchical models and latent variable models.
- **Uncertainty** is the key of statistics, and Bayesian approaches **deepen your understanding** of uncertainty

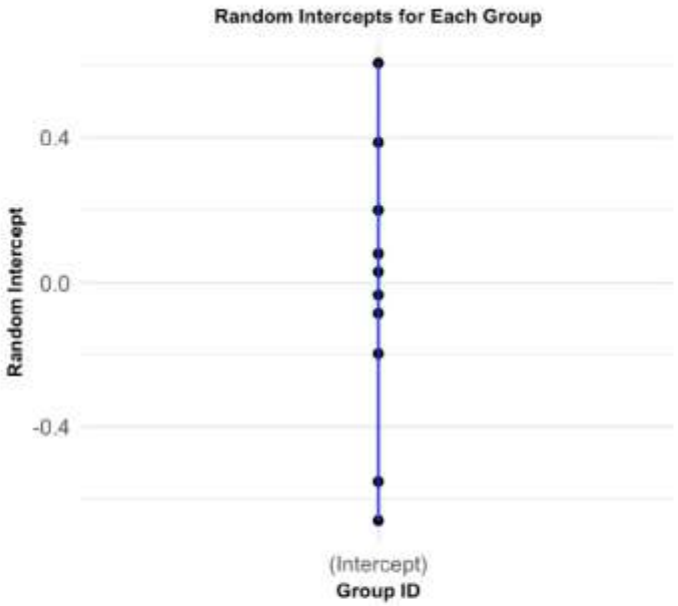


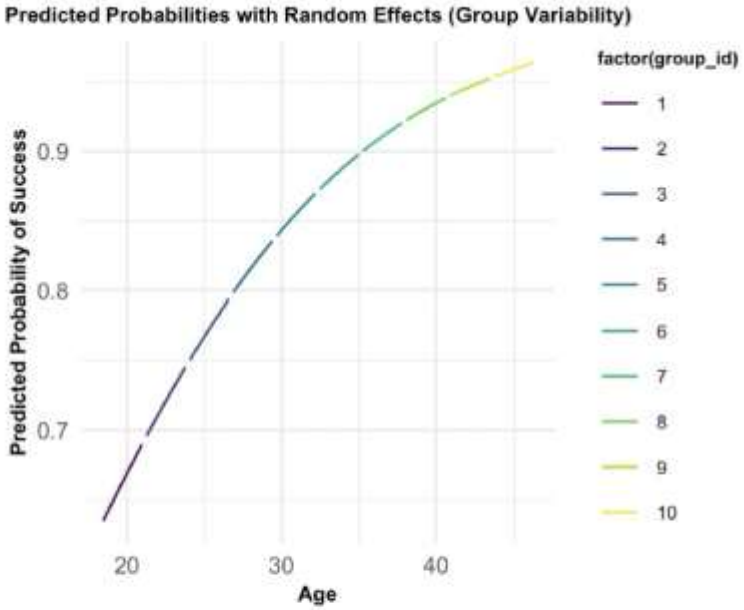
你学到什么？

1. 遗传学和基因组学基本概念
2. 遗传学中使用的统计模型/假设检验
3. 全基因组关联分析（GWAS）
4. 全转录组关联分析（TWAS）
5. 孟德尔随机（MR）
6. 单细胞/空间转录组分析



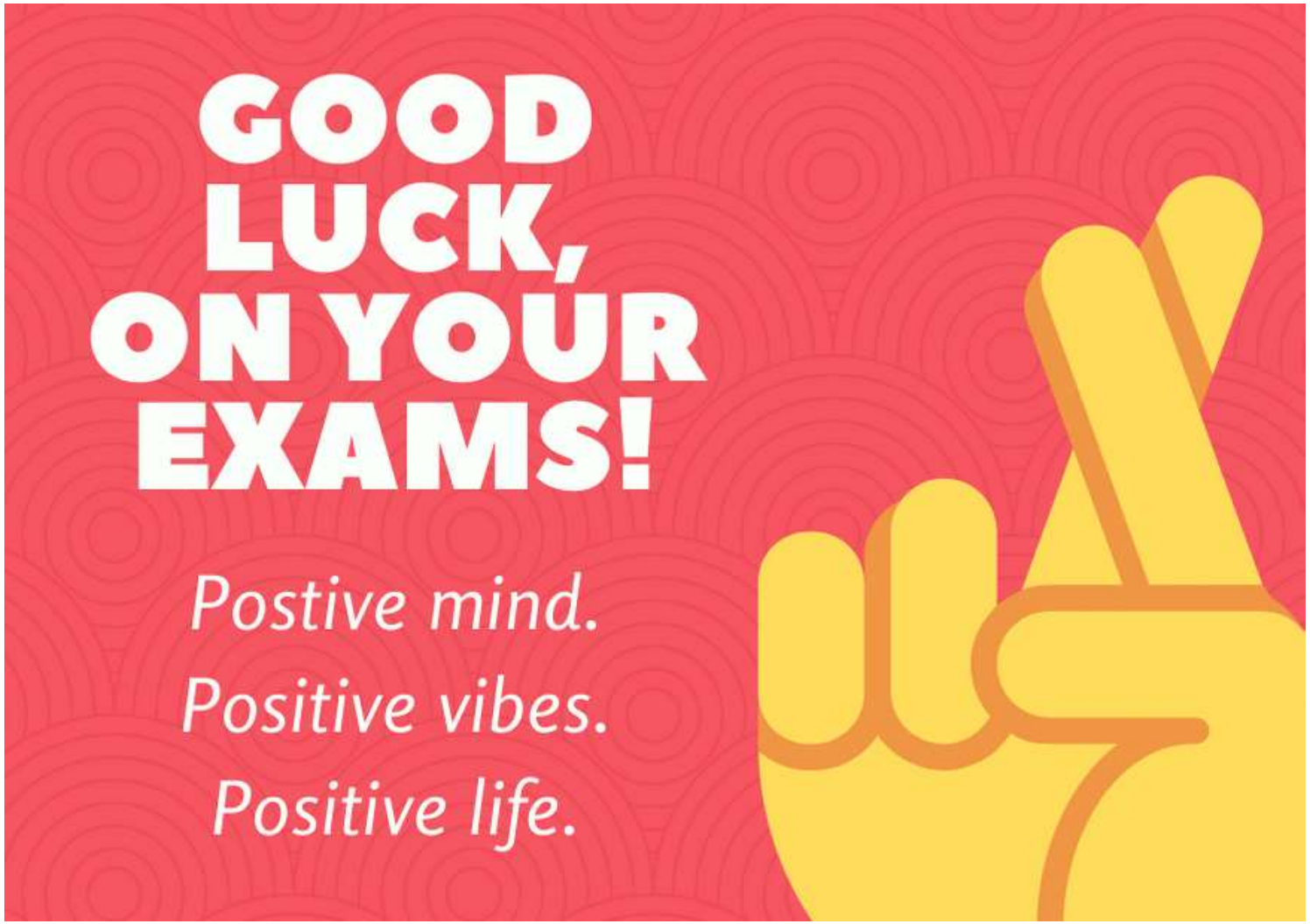






STA4102 Generalized Linear Models

- Why GLM?
  - Model non-Gaussian distributed data, with mean-variance dependence
    - Binary: experiment success/failure
    - Count: number of events (e.g., gene expression)
    - Nonnegative continuous: patients survival time, insurance claims
  - Model nonlinearity relationship between outcome and predictors
  - Address correlations
    - Clustered/longitudinal data: repeated measurements for the same unit
- Example GLMs: Logistic Regression, Poisson/negative binomial Regression, Gamma Regression etc.


GOOD LUCK, ON YOUR EXAMS!
Postive mind.
Positive vibes.
Positive life.